Switzer

INSTRUCTION MANUAL

FOR

PRESSURE SWITCHES
AND
DIFFERENTIAL PRESSURE SWITCHES

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OPERATION AND MAINTENANCE INSTRUCTIONS

COMMON FOR ALL PRESSURE SWITCHES / DIFFERENTIAL PRESSURE SWITCHES

SWITZER Pressure Switch is a simple electro mechanical device operating on basic principles of Levers and opposing forces. Three essential elements, various combinations of which form the basics for presenting hundreds of models to suit a variety of industrial applications. They are:

1. sensing element either of bellows, bourdon or diaphragm (metallic or elastomeric)
2. a stable spring to define the range and
3. a snap acting microswitch available in a wide variety.

OPERATIONS

Process pressure when applied to the sensing element creates a force which when overcomes that of a pre-tensioned spring, moves a balancing arm to effect in turn the minimal movement required to actuate a microswitch(s).

MAINTENANCE

Instruments are designed with rugged components that it seldom requires maintenance. Occasional cleaning of the moving parts, checking of the microswitch(s) and ensuring firm electrical contacts at the terminals will provide a trouble-free long performance.

GENERAL PRECAUTIONS

Ensure that

1. Correct positioning of gaskets when covers are fixed.
2. Electrical entries and cables are properly sealed with correct cable gland. If in doubt, consult factory.
3. Process pressure never exceeds declared maximum working pressure.
4. Connected electrical load never exceeds declared maximum electrical capacity both in ampheres and volts.
5. The instrument housing is never held to tighten at on the process line. A hex. or square provided on the sensing element housing should be held by a spanner for tightening.
6. A master gauge of accuracy more than 0.5% is to be used for setting actuating point as the setting accuracy will be the same as that of the master gauge used.
7. The lock screws are loosened before the range spring screw is operated and tightened again after set point is achieved.
8. The instrument is mounted rigidly, to avoid shock and vibration.
9. Out door installations are with sufficient protections against aggressive air dust, temperature and water.
10. Suitable dampner/snubber to be used in rapidly fluctuating pressure lines.
11. A condensate coil should be used invariably for steam service.
12. All screws are tight.

ADDITIONAL INSTRUCTIONS

MODEL 021

Description : Specially contoured synthetic elastomer diaphragm constitutes the sensing element. Due to its elastomeric nature, over pressure beyond the specified value will permanently damage the diaphragm needing replacement.

Instructions : This model provides fixed differential value, which is different for different microswitch codes. As the instrument is mainly intended for low pressure, ensure that the pressure source is connected without any leak and that it does not produce pressures higher than the maximum working pressure. This would ensure avoiding application of inadvertant high pressures. Switch setting is achieved by adjusting the range spring when process line is connected to the pressure source and master gauge. A hand operated pressure source is recommended for this instrument. Avoid using reduced compressed air pressure source.

MODEL 201

Description : This model provides fixed differential value, which is different for different microswitch codes '0', '2' and '3' even though the range of the pressure switch is same. However the values claimed in literature or offer are nominal and are subject to variation due to design constraints.

Instructions : To adjust the switch setting, remove the front cover, unlock the screw holding the lock plate clamping the range screw. Rotate the range

suitably to select the switching point either for fall in pressure or rise in pressure setting.

MODEL 202

Description : This model provides an adjustment of the ON–OFF Differential for a narrow band value. This is achieved by using a special microswitch.

Instructions : The special microswitch has a knob by rotating which from minimum to maximum position, the ON–OFF Differential is varied from the minimum to maximum specified value. Empherical marking from 'A' to 'F' are provided on the switch knob to enable repeative setting. However the values claimed in literature or offer are nominal and are subject to variation due to design constraints.

MODEL 203 / 303

Description : ON–OFF Differential value can be adjusted in a wider manner from above 10 to 15% of the span to a maximum of 60 to 80% as specified against each range. These values will vary with different switch combinations. This facility is achieved by employing an auxiliary spring brought into action when the switch actuating plate moves up and operates a microswitch. Variations in the tension of the spring decides the pressure difference between the on point and off point.

Instructions : Loosen the knob above the auxiliary spring provided at the right side (while facing the instrument) will ensure basic minimum differential. Adjust the range spring (located at the left side) first to decide the actuating point. Load the auxiliary spring by turning the knob clockwise and observe the actuating point in the master gauge. Stop the rotation until the desired activating point (on point) is achieved. Bring down the nyloc screw and tighten it against the knob to ensure locking.

Precaution : The switch actuating screw on the balancing arm is critically adjusted. Disturbance of this would result in not achieving the desired result while adjustments are made. To reset the micro switch for correct operation adjust the height of the actuating screw such that the balancing arm is not in contact with the auxiliary spring seat at the time of switch deactuation. This alone will ensure unloaded condition of the auxiliary spring during deactuation.

MODEL 204, 208 & 209

Description : This diaphragm operated instrument is capable of withstanding high static pressure without affecting its operating values. This is achieved by using a specially contoured SS diaphragm held firmly against a back plate exactly matching its contour. This prevents the distortion of the diaphragm against applications of over pressure.

Instructions : For trouble-free repeative operation, the diaphragm contour should not be disturbed during it entire life span. If an attempt is made to remove the diaphragm and repositioned, even a minute misalignment would result in diaphragm distortion and consequent misoperations. Diaphragms are to be factory replaced only and customer replacement is not recommended. However cleaning of the diaphragm chamber can be performed by flushing in a proper cleaning fluid which is compatible with SS and the "O" ring material.

MODEL 205

Description : This instrument has an interdependent high and low set point facility. Two microswitches are provided for Hi and Lo. While microswitch for low set point is actuated by the common range spring, the microswitch for the high set point is actuated by the balancing arm after it overcomes an additional spring force created by the auxiliary spring setting located at the right side (while facing the instrument). First release the tension on the auxiliary spring by rotating the knob above it anticlockwise. Adjust the low set point by adjusting the range spring against master gauge. Load the auxiliary spring by rotating it anticlockwise to required tension and observe the master gauge for actuation of the high set point microswitch. Bring down the nyloc screw and tighten it against the knob to ensure locking.

MODEL 281

Description : This instrument provides an independently adjustable high and low set point facility. A single sensing element actuates two different balance arms, through an unique pattented floating arm. Two sets of range springs, range scales, balance arms and microswitches are independently operated.

Instructions : Both the range springs are clearly marked for high and low functions. First set the low range spring and adjust the desired value for the actuation of the microswitch against a master gauge. The high spring can then be adjusted in a similar way to adjust the desired high set point against a master gauge. Ensure that the correct microswitch is connected while range settings are done.

MODEL 301

Description : This pressure differential switch employs a single specially contoured 316 LSS diaphragm as the sensing element. High and low pressure are applied on either side of the same diaphragm and this design feature straightaway eliminates errors due to the difference in area, a common problem present in twin element pressure differential switches. The task of transferring the resultant movement of the diaphragm is achieved by employing an unique patented motion transfer assembly which is made sensitive to minute movements of the diaphragm but insensitive to application of very high static pressures. The diaphragm is held between top and bottom chambers of low and high pressures respectively.

Instructions : Connect the pressure source to the high port venting the low port. Set the range screw for switch operation against a master gauge connected to the same line. Do not attempt dismantling the motion transfer mechanism and the diaphragm housing as it would permanently disturb the factory settings. Special equipments are needed to reassemble and hence reassembly is not recommended at the user end. However, cleaning of the diaphragm chamber can be performed by flushing with a proper cleaning fluid which is compatible with SS and the 'O' ring material.

MODEL 306

Description : This differential pressure switch employs a single specially contoured synthetic elastomer diaphragm.

High and low pressures are applied on either side of the diaphragm and this design feature straight away eliminates errors due to difference in area, a common problem present in twin element pressure differential switches. The task of transferring the resultant movement of the differential is achieved by employing an additional sealing diaphragm above the low pressure chamber.

Instructions : Connect the pressure source to the high port venting the low port. Set the range screw for switch operation against a master gauge connected to the same line. Do not attempt dismantling the diaphragm housing as it would permanently disturb the factory settings. Special equipments are needed for reassembly and hence it is not recommended at the user end. However, cleaning of the diaphragm chamber can be performed by flushing in a proper cleaning fluid which is compatible with SS and the diaphragm material. Avoid using reduced compressor air pressure source.

MODEL 310

Description : This pressure differential switch employs a synthetic rubber diaphragm (Nitrile, Neoprene or Silicon as ordered) as the sensing element. High and low pressures are applied on either side of the same diaphragm and this design feature eliminate errors due to the difference in area of diaphragms, a common problem present in twin element pressure differential switches. The result in movement of the diaphragm is transferred via 'O' ring shaft seal to a actuating lever which operates the microswitch. The diaphragm is held between top and bottom chambers of low and high pressures respectively.

Instructions : Connect the pressure source to high port venting the low port. Set the range screw for switch operation against a master gauge connected to the same line. Do not attempt dismanthing the motion transfer mechanism and the diaphragm housing as it would permanently disturb the factory settings. Special equipments are needed to reassemble and hence reassembly is not recommended at the user end. However, cleaning of the diaphragm chamber can be performed by flushing with a proper cleaning fluid which is compatible with diaphragm and the 'O' ring material.

MODEL 313

Description : This differential pressure switch is similar to Model 310, but with adjustable on and off differential facility. Adjustments can be made in a wider manner from about 10–15% of the span (minimum) to a maximum of about 60–80% as specified in literature against each range. This facility is achieved by employing a auxiliary spring brought into action when the switch actuating plate moves up and operates a microswitch. Variations in the tension of the spring decides the pressure difference between the on point and off point.

Instructions : Loosen the knob above the auxiliary spring provided at the right side (while facing the instrument) will ensure basic minimum differential. Adjust the range spring (located at the left side) first to decide the actuating point. Load the auxiliary spring by turning the knob clockwise and observe the actuating point in the master gauge. Stop the rotation until the desired activating point (on point) is achieved. Bring down the nyloc screw and tighten it against the knob to ensure locking.

Precaution : Please refer to Model 203 / 303 precautions.